

Special Issue on Silent Data Corruptions— From Silicon to Cloud Data Centers and AI Systems of Huge Scale

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Silicon chips and the computing systems they power are not perfect—including all layers of the hardware and software stacks. Failures can happen where chips are deployed; more chips means more failures.

A *defect* is the real physical mechanism behind a failure, and it can be either internal (material and process imperfections or wear-out) or external (radiation). A *fault* is a model that describes the practically infinite universe of defects; fault models are used to quantify the methods that dealing with defects. An *error* is the manifestation of a fault (or a defect that it models) in any abstraction layer. A *failure* of a computing system is a deviation of the delivered service from the expected one.

In large-scale computing, that is, data centers and high-performance computing (HPC) systems, the *failure rate*, i.e., the number of failures happening over time [one FIT (failures in time) means that one failure happens per billion hours of operation], is an important metric. However, not all failures are the same.

When a failure is obvious or visible (i.e., detected by hardware, software, or a human), the system operators can take action and, with a small or large effort, return the system to correct operation. Even if no recovery action can be performed, it is useful to know that the system fails and to stop using it. However, a *silent data corruption* (SDC) (or silent error or silent data error) is a failure that is not detected by any hardware or software mechanism (or is detected only after severely affecting the computing system service). SDCs have been a long-known effect of defects or radiation on silicon: the result of a computation is wrong, but no hardware or software mechanism has been provided

or was able to detect it—the computing system is fully operational, and no exception was raised, nothing unexpected was logged or observed. This definition of SDC itself points to something extremely rare. Given the engineering effort and the hardware and software resources that the computing industry invests during design, manufacturing, and operational time to minimize the impact of silicon defects, it has always been a common belief that silicon chips either work correctly or malfunction grossly and need to be replaced.

Failures in the machines that our lives depend so much on are rarely acceptable, but at least when noticed, measures can be taken. When the failure is silent, *zero* is the only acceptable rate.

For decades, the common belief was that SDCs were a myth, or, in a worst-case scenario, an SDC might affect the unlucky user of one in a million (or billion) computers. Since early 2021, hyperscalers have told us that the reality is different (we can safely assume that the problem was known and under investigation at least two or three years earlier); about 1 chip in a 1000 produces SDCs. [Table 1](#) summarizes public disclosures about SDCs at scale during the last five years, including the date of disclosure, medium, and type of chips and systems that the report was about.

The disclosures were followed by actions from industry (individually or collectively) and the research community. [Table 2](#) summarizes the initiatives taken, including collaborative research as well as dissemination activities.

This special issue of *IEEE Micro* on SDCs is the latest activity on the topic and takes a detailed snapshot of the state of the art in the five years since the first public disclosures (and probably a little less than a decade from the first private reports from hyperscalers to chip vendors).

The special issue aims to serve as a reference point (and for beginners, as a single starting point) on the topic of SDCs. In particular, the special issue seeks answers to the following (non-exclusive list of) questions:

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APPENDIX: RELATED ARTICLES

- A1. N. Saxena, S. Hukerikar, and S. Hari, "Silent data corruption: Optimal mitigation strategies for data center computing," *IEEE Micro*, vol. 46, no. 1, pp. 10–18, Jan./Feb. 2026, doi: [10.1109/MM.2025.3643799](https://doi.org/10.1109/MM.2025.3643799).
- A2. N. Karystinos, G.-M. Fragkoulis, O. Chatzopoulos, D. Gizopoulos, and S. Gurumurthi, "Harpocrates⁺⁺: Automated functional program generation against CPU faults and silent data corruptions," *IEEE Micro*, vol. 46, no. 1, pp. 19–28, Jan./Feb. 2026, doi: [10.1109/MM.2025.3640385](https://doi.org/10.1109/MM.2025.3640385).
- A3. P. Bose et al., "Robust power management: A key aspect of silent data corruption-free systems," *IEEE Micro*, vol. 46, no. 1, pp. 29–33, Jan./Feb. 2026, doi: [10.1109/MM.2025.3644998](https://doi.org/10.1109/MM.2025.3644998).
- A4. M. Shamsa, D. Lerner, A. Sinha, T. Macieira, and Y. Varma, "Methodical detection of silent data errors from foundry to fleet," *IEEE Micro*, vol. 46, no. 1, pp. 34–43, Jan./Feb. 2026, doi: [10.1109/MM.2025.3646615](https://doi.org/10.1109/MM.2025.3646615).
- A5. H. Pei, L. Lausen, and G. Karypis, "Connecting the impact of silent data corruption with different training characteristics: An empirical study," *IEEE Micro*, vol. 46, no. 1, pp. 44–51, Jan./Feb. 2026, doi: [10.1109/MM.2025.3642709](https://doi.org/10.1109/MM.2025.3642709).
- A6. C. Vallin, R. Bhimanadhuni, and R. S. Chappell, "Silent data corruption challenges in modern AI public cloud deployments," *IEEE Micro*, vol. 46, no. 1, pp. 52–54, Jan./Feb. 2026, doi: [10.1109/MM.2025.3641148](https://doi.org/10.1109/MM.2025.3641148).
- A7. O. Chatzopoulos, M. Trakosa, H. D. Dixit, S. Sankar, and D. Gizopoulos, "Phoebe: Measuring the unmeasurable—Demystifying silent data corruptions in AI accelerators through microarchitectural modeling," *IEEE Micro*, vol. 46, no. 1, pp. 55–65, Jan./Feb. 2026, doi: [10.1109/MM.2025.3646859](https://doi.org/10.1109/MM.2025.3646859).
- A8. N. George et al., "Silent data corruption in artificial intelligence: A growing challenge for large-scale machine learning," *IEEE Micro*, vol. 46, no. 1, pp. 66–72, Jan./Feb. 2026, doi: [10.1109/MM.2025.3645670](https://doi.org/10.1109/MM.2025.3645670).

TABLE 1. Public discussions of SDCs (2021–2025).

Who	When	Where	SDCs in
Meta	February 2021	https://arxiv.org/	Cloud CPUs
Google	June 2021	HotOS	Cloud CPUs
Google	June 2023	ISCA	DNN training and TPUs
Alibaba	October 2023	SOSP	Cloud CPUs
Meta	April 2024	ASPLOS	DNN inference
DeepSeek	November 2024	SC	LLM training and GPUs
Amazon	February 2025	https://arxiv.org/	LLM training and GPUs
Tesla	June 2025	X post	ML training chips

ISCA: International Symposium on Computer Architecture; ASPLOS: ACM International Conference on Architectural Support for Programming Languages and Operating Systems; TPUs: tensor processing units; DNN: deep neural network; ML: machine learning; LLM: large language model; SOSP: Symposium on Operating Systems Principles; SC: Supercomputing Conference; HotOS: Workshop on Hot Topics in Operating Systems.

- How severe is the problem of SDCs?
- Which types of computing chips (CPUs, GPUs, and domain-specific accelerators) are more likely to produce SDCs?

- What is an SDC at different domains [cloud, HPC, and artificial intelligence (AI) training/inference]?
- Which workload types are more likely to produce SDCs?
- Are new types of silicon defects or known ones responsible for SDCs?
- What kind of circuit, microarchitectural, or architectural techniques can reduce SDCs?
- What kind of compilation and system software techniques can reduce SDCs?
- At what level of abstraction is SDC mitigation more cost-effective?
- What can chip vendors do to mitigate SDCs?
- What can foundries and the electronic design automation industry do to mitigate SDCs?
- What can hyperscalers do to minimize the likelihood of SDCs in customer workloads?

A dream team of contributors to the special issue [with industry authors from AMD, Meta, Amazon, Nvidia, Microsoft, IBM, Intel, Google, and Arm, either in individual articles or through their role in the Open Compute Project (OCP) Resilience workstream] sheds light on different aspects of the problem and answers some of the aforementioned questions.

The main conclusions that can be drawn (directly or indirectly) when readers go through the pages of the eight articles are the following:

- › The problem of SDCs severely impacts all compute chips: CPUs, GPUs, and AI accelerators (AIAs).
- › Cloud and HPC computing, due to their increasing scale, are very sensitive to the problem of SDCs.
- › Data parallel architectures (GPUs, TPUs, are AIAs) are likely suffering more SDCs than control architectures (CPUs).
- › AI workloads and systems are more difficult to analyze for SDCs. The definition of an SDC itself is a challenge for AI workloads.
- › Modeling and simulation at different layers (circuit, microarchitecture, and architecture) are crucial in measuring and mitigating SDCs.
- › SDCs mitigation needs strategies at all stages of design, manufacturing, testing, deployment, and operation.

As always, the design of large-scale computing systems is a complex problem of *optimization* and *space exploration* where the most important parameters compete with each other: performance,

power/energy consumption, resilience, yield, and cost. Different systems are optimized for different parameters. When resilience to faults (and SDCs avoidance) is important, any solution comes with a cost. The question for SDCs mitigation is where the cost should go to. Should some performance be sacrificed? Should the final system be more expensive?

*SDCs MITIGATION NEEDS STRATEGIES
AT ALL STAGES OF DESIGN,
MANUFACTURING, TESTING,
DEPLOYMENT, AND OPERATION.*

Should it consume some more energy? “All of the above” is the most likely answer.

Most probably, almost a decade after the problem of SDCs started escalating, we now have more questions than answers. However, this is expected because computing systems’ scale, complexity, and density keep increasing (nothing new!), and the challenges for quality and correctness will always be at the forefront of computer science and engineering. Hopefully, with the investments from all major players in dealing with the problem, we will soon be informed that SDC-producing CPU, GPU, and AIA chips will be orders of magnitude less than the terrifying, “one in a thousand” that everyone has reported during the last few years.

The articles of the special issue cover a wide spectrum of aspects related to SDCs: diverse types of compute chips, a broad range of large-scale system installations, different detection and mitigation approaches at the hardware or the software layers, and discussions about the role of modeling and simulation as well as metrics and measurements. [Table 3](#) summarizes the aspects that each of the articles touches upon. Most articles, of course, discuss multiple aspects of the SDCs problem.

Saxena et al.^{A1} advocate for optimal mitigation strategies in data centers and the need to complement error correction codes with algorithm-based error detection. Karystinos et al.^{A2} present the state-of-the-art automated method, which generates functional test programs and grades them using the gem5 microarchitectural simulator; the programs detect defects in real x86 CPUs. Bose et al.^{A3} discuss the tight connection between diligent design and characterization for power, droops, margins, and the problem of SDCs. Shamsa et al.^{A4} discuss marginal defects characterization and improvements of different test stages.

TABLE 2. SDCs research and dissemination initiatives.

Who	When	What
Meta	June 2022	SDCs RFP awards five universities
AMD	October 2023	SDC’s collaboration with two universities
OCP	June 2024	SDC’s research awards to six universities
IEEE Computer Society	June 2024	First RAS Summit (second edition coming in 2026)
IEEE Computer magazine	June 2025	General readership article
IEEE Computer Society and Association for Computing Machinery	2023–2025	Tutorials, special sessions, and panels at MICRO, ISCA, HPCA, ITC, VTS, and IOLTS
IEEE Micro	2026	This special issue on SDCs

RAS: Reliability, Availability and Serviceability; OCP: Open Compute Project; RFP: request for proposal; MICRO: International Symposium on Microarchitecture; ISCA: International Symposium on Computer Architecture; HPCA: International Symposium on High-Performance Computer Architecture; ITC: International Test Conference; VTS: VLSI Test Symposium; IOLTS: International Symposium on On-Line Testing and Robust System Design.

TABLE 3. SDCs aspects covered by the special issue articles.

Aspect of SDCs	Articles
CPUs	Karystinos et al. ^{A2} Bose et al. ^{A3} and Shamsa et al. ^{A4}
GPUs	Saxena et al. ^{A1} Pei et al. ^{A5} and Vallin et al. ^{A6}
AIAs	Chatzopoulos et al. ^{A7} and George et al. ^{A8}
Testing	Karystinos et al. ^{A2} Bose et al. ^{A3} and Shamsa et al. ^{A4}
Power management	Bose et al. ^{A3}
Inference	Saxena et al. ^{A1} Chatzopoulos et al. ^{A7} and George et al. ^{A8}
Training	Saxena et al. ^{A1} Pei et al. ^{A5} Vallin et al. ^{A6} and George et al. ^{A8}
Simulation	Karystinos et al. ^{A2} Bose et al. ^{A3} and Chatzopoulos et al. ^{A7}
Measurement and metrics	Saxena et al. ^{A1} Karystinos et al. ^{A2} Shamsa et al. ^{A4} Chatzopoulos et al. ^{A7} and George et al. ^{A8}
Injection	Saxena et al. ^{A1} Karystinos et al. ^{A2} and Chatzopoulos et al. ^{A7}
Public cloud	Vallin et al. ^{A6}
Data center	Saxena et al. ^{A1} Karystinos et al. ^{A2} Shamsa et al. ^{A4} Vallin et al. ^{A6} Chatzopoulos et al. ^{A7} and George et al. ^{A8}
Mitigation	Saxena et al. ^{A1} Karystinos et al. ^{A2} Shamsa et al. ^{A4} and George et al. ^{A8}
Call to action	George et al. ^{A8}

The remaining four articles discuss SDCs in the AI chips and systems domains. Pei et al.^{A5} present the impact of defective chips in large language model training and how different training knobs change it. Vallin et al.^{A6} discuss the challenges that large AI public cloud setups face due to SDCs. Chatzopoulos et al.^{A7} present the first comprehensive full-system microarchitecture-level SDC analysis framework for AI accelerators and the benefits it brings to design hardware- or software-based resilience. Finally, George et al.^{A8} bring a consolidated industry view (from OCP member companies) on how SDCs impact large-scale machine learning systems, advocate for industry-academia collaboration, and point to different mitigation approaches for the problem.

My appreciation goes to all the authors of the accepted and rejected articles who shared their insights on the topic and prepared (and revised) their articles.

The reviewers did an outstanding job providing valuable feedback and working on very tight schedules. The *IEEE Micro* staff was very responsive to all my requests about status updates and corresponding actions.

Last but not least, I wish to warmly thank *IEEE Micro* Editor-in-Chief Dr. Hsien-Hsin “Sean” Lee, who immediately embraced and welcomed my proposal for this special issue. His guidance and assistance throughout the process was instrumental. Thank you very much, Sean.

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